

Helper Functions API

ASTR 3070 Fall 2025

fetch_filenames

`fetch_filenames(dir)`

Fetches a list of filenames from a directory.

Args:

`dir (str)`: path of the file directory

Returns:

`list[str]`: list of filenames

stack_master

`stack_master(files)`

Stacks frames to create a master file.

This does NOT align the frames when stacking.

Args:

`files (list)`: list of filenames

Returns:

`(np.array)`: final image stack in the form of a 2D array

calibrate_lights

`calibrate_lights(light, dark, flat_bias)`

Applies dark and bias/flat corrections to individual light frames.

Args:

`light` (list[str]): list of light frames

`dark` (np.array): master dark frame

`flat_bias` (np.array): master flat/bias frame

Returns:

list[np.array]: list of corrected light frames

align_image

`align_image(list_of_images1, list_of_images2, reference_image)`

Aligns images from two nights.

- To use with one night, use the same list in both parameters and divide the returned array by 2
- To use with multiple nights, use the returned array as one parameter and the next night of images as the second, then repeat as needed

Args:

`list_of_images1` (np.array): images from first night

`list_of_images2` (np.array): images from second night

`reference_image` (np.array): image to use as reference for alignment

Returns:

(np.array): aligned master stack array

create_table

```
create_table(pos_filename, pho_filename, sep_pos = '\s{2,}', sep_mag = '\s+')
```

Create a table from .pos and .pho files.

Args:

pos_filename (str): given .pos (coordinates) file

pho_filename (str): given .pho (magnitudes) file

sep_pos (str, optional): modify if the data is not separated correctly

sep_mag (str, optional): modify if the data is not separated correctly

Returns:

(np.array): aligned master stack array